A one-parameter crack detector on SEM micrographs, and the pixel budget it spends

DRAFT — not submitted. Every number is generated from measurement files by paper/make_paper.py; nothing in this document is typed by hand twice.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on the only public crack data we could find in its own modality and material class: 31 whole scanning-electron fields of Inconel 718 with pixel-exact masks.

The headline result is that its one constant did not need refitting. mad_k, fitted years earlier on SEM steel in another laboratory, is the best of seven values tested on this independent alloy, and the curve falls away monotonically on either side.

We report the accuracy primarily as a pixel budget rather than as F1 or IoU. On an image that is 1.80% crack, F1 and IoU discard the 98% of pixels that are correctly left alone, and they cannot state the quantity this tool actually outputs — an area fraction. The detector marks 2.34% of the image where the human marked 1.80%: of that, 0.78% is right and 1.56% is wrong, so an area fraction read off this tool is 1.30 times too high. Accuracy is 97.42%, which is worse than marking nothing; balanced accuracy is 69.82% and MCC 0.3402. In the conventional units it is F1 0.3444 and IoU 0.2137, 9.8 times the F1 of marking every pixel.

Two results cut against the method and we report them rather than omit them: against seven classical detectors tuned identically it ranks third of seven, within 0.0186 F1 of the leader; and cropping these same images around their cracks — as the dataset's own test split does — moves the reported area fraction from 1.30x to 0.78x and F1 from 0.3444 to 0.4958, with no change to the detector at all.

Status of this draft. This is one alloy and 31 images. Section 7 says what that is and is not enough for.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales, takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. It has one free constant, mad_k. There is no training step and no model file.

Reporting such a thing honestly turns out to be harder than building it, and most of this paper is about the reporting. The detector is a threshold; what is less obvious is that the scores conventionally used to grade one — F1 and IoU — cannot express the error a materials laboratory actually suffers, which is a crack area fraction that comes out too high or too low by some factor. Section 3.3 reports that factor, and Appendix A gives it for every set we measured. It ranges from 0.74x to 76.65x for the same unchanged detector.

2 The detector

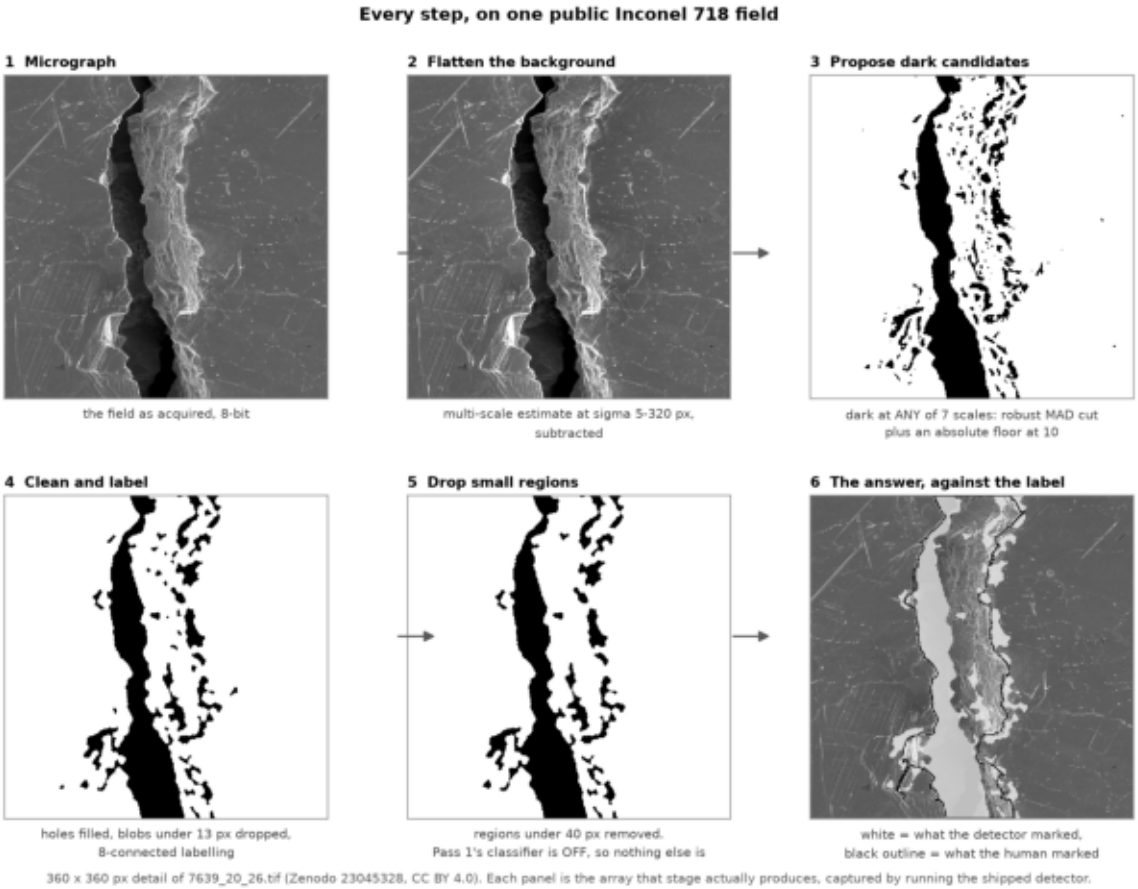


Figure 1. P6 - pipeline in pictures

Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup. There is no training step and no model file, and every panel in Figure 1 is the array that stage actually produces — captured by running the shipped code, not drawn.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric with no free parameters: paired median IoU +0.2296 over thirteen specimens, p = 0.0024. Three specimens that had scored recall 0.0000 at every candidate setting ever tried were never invisible to the proposal stage; the classifier was rejecting them.

Tiles cannot stand in for frames. Rules validated on 512 px crops deleted 40–43% of hand-marked crack when run on whole frames. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is extrapolating on the object it most needs to keep.

3 On public SEM data with pixel-exact ground truth

One set: the Inconel 718 crack class of a public cross-material microstructure collection (Zenodo 23045328, CC BY 4.0), 31 whole 1024 px fields of a nickel superalloy at 1.80% crack prevalence. It is the only public crack segmentation data we found in this modality and material class; Section 7 records what else was searched and rejected.

Scoring is at 0 px tolerance with mad_k chosen leave-one-image-out. Two properties of the

data had to be established before any score could be trusted: the masks are anti-aliased binary rather than strictly two-valued, so the scorer checks bimodality and refuses a mask that fails it; and the images are 16-bit TIFFs, where an ordinary 8-bit conversion drove recall to 0.0012 before it was caught. Taking the high byte reproduces the dataset authors' own 8-bit PNGs exactly, so that is the conversion used.

We report whole fields rather than the collection's released crops. Every one of the 18 source images behind those crops also appears among the originals, so the two are not independent sets, and the crops are cut around their cracks — which raises prevalence from 1.80% to 4.53% and the score with it, with no change to the detector.

3.1 What it scores

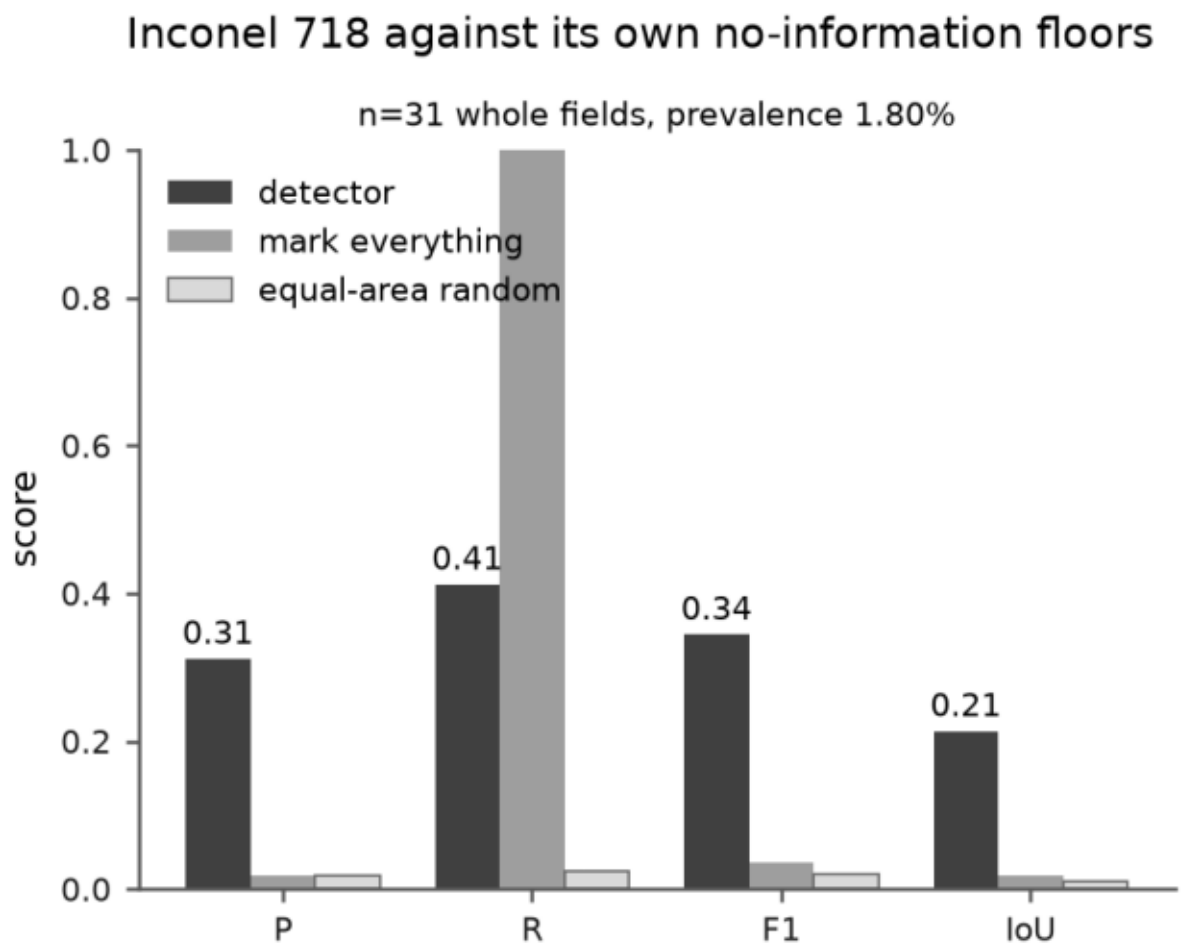


Figure 2. P1 - public scores

F1 0.3444 and IoU 0.2137, against 9.8 times the F1 and 11.8 times the IoU of marking every pixel. A mask of the same area with its pixels scattered at random scores F1 0.0196.

3.2 The one free constant, unrefitted

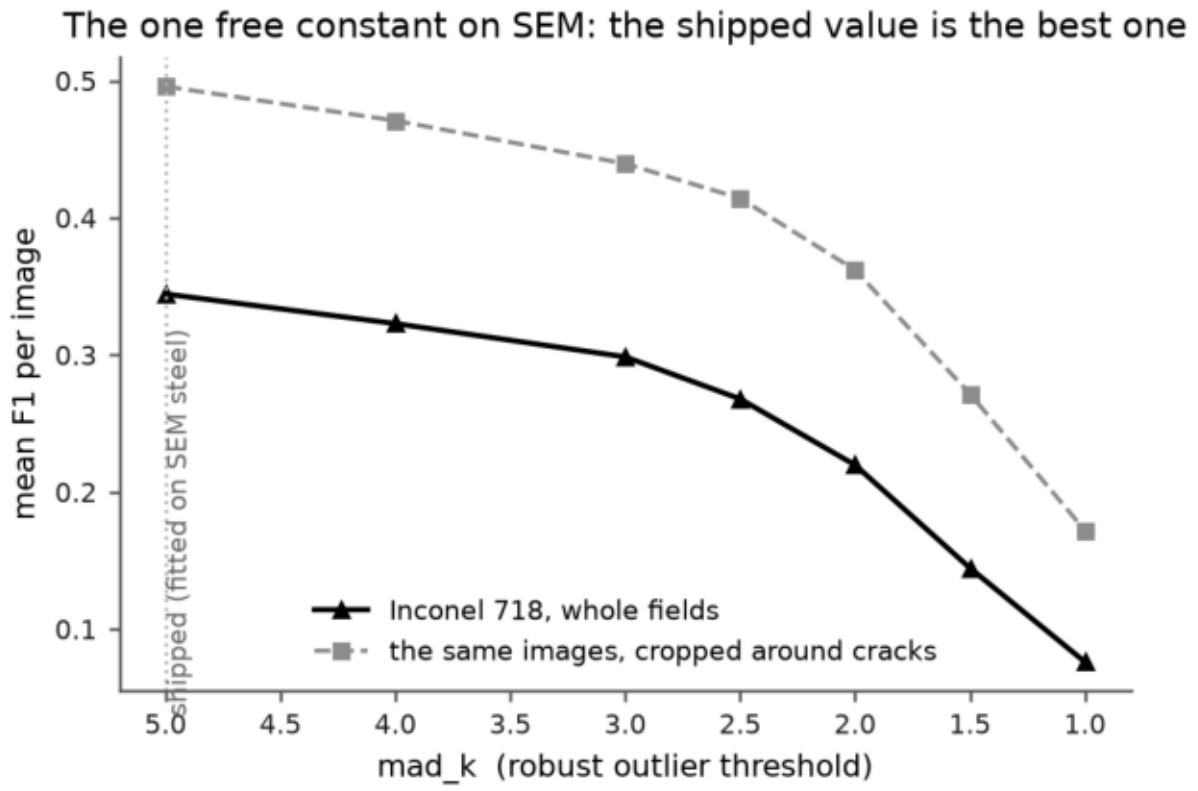


Figure 3. P2 - operating curve

mad_k = 5.0 was fitted on SEM steel in a different laboratory, on a different alloy, years before this dataset existed. On these whole fields it is the best of the seven values tested, and the curve falls away monotonically below it.

That is the strongest result in this paper. A detector with one free parameter, carried unchanged to an independent laboratory's alloy, lands on that laboratory's optimum. The dashed series is the same images cropped around their cracks, which shifts the whole curve up by about 0.15 F1 without moving its peak — the framing changes the score, not the setting.

3.3 Scored as a pixel budget

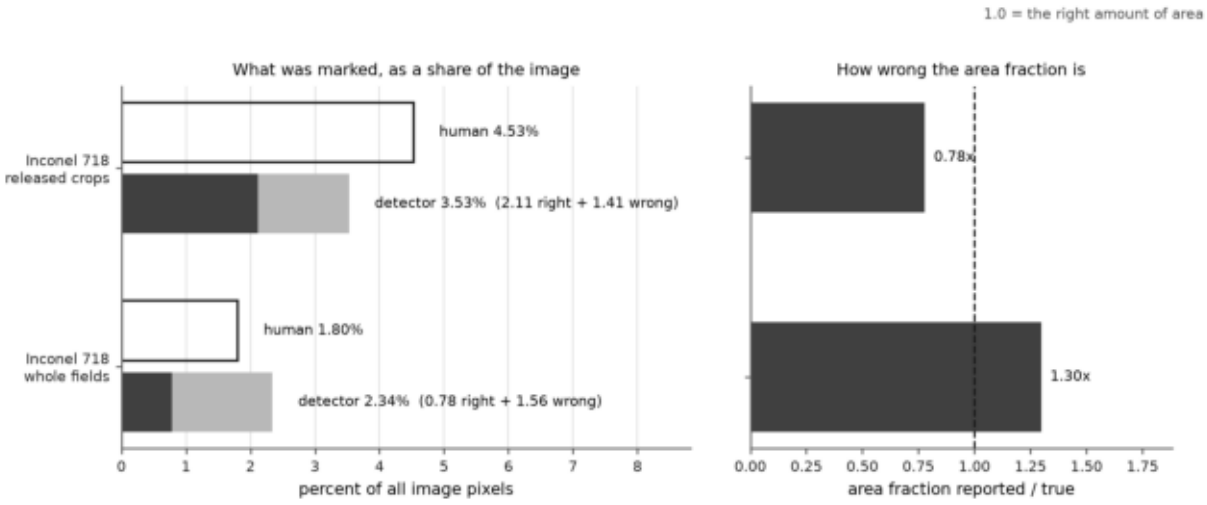


Figure 4. P5 - pixel budget

Precision, recall, F1 and IoU all discard the true negatives. On an image that is 1.80% crack that is 98% of the evidence, and it lets two very different detectors share a score — one marking 0.3% of a frame and one marking 5% can land on the same IoU. The quantity this tool reports to a user is an area fraction, a pixel percentage, so the scores belong in the same units as the answer.

	whole fields	released crops
labelled crack	1.80%	4.53%
detector marked	2.34%	3.53%
of which correct	0.78%	2.11%
of which wrong	1.56%	1.41%
area fraction reported / true	1.30x	0.78x
accuracy	97.42%	96.17%
balanced accuracy	69.82%	72.09%
MCC	0.3402	0.4840

Three things only this view shows.

The area fraction a user reads off this tool is about 1.30 times too high on whole fields. That is the error that reaches a materials conclusion, and no F1 or IoU states it.

The bias reverses between the two framings of the same images — 1.30x on whole fields, 0.78x on the crops cut from them. The detector did not change.

Accuracy is 97.42% and is worthless. Marking nothing at all would score 98.20%. It is printed here only so that it is never quoted: at this prevalence the measures worth reading are balanced accuracy and MCC, and those are 69.82% and 0.3402.

3.4 Against classical detectors

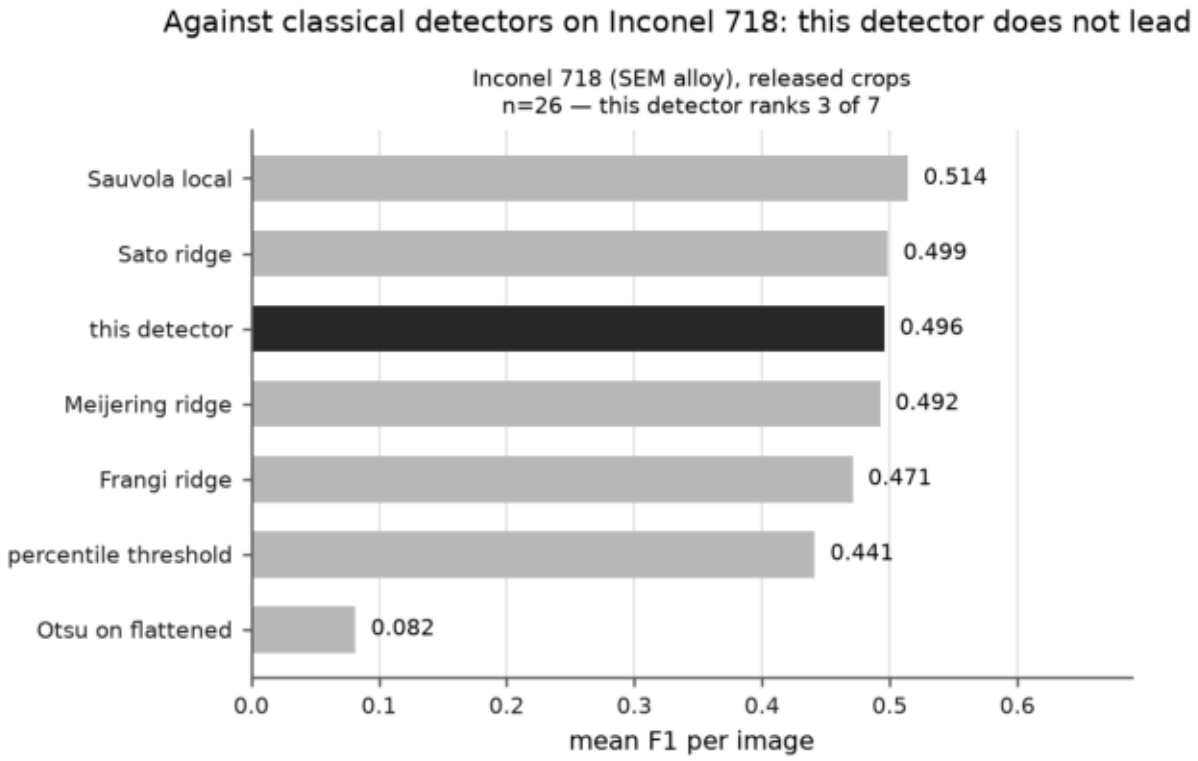


Figure 5. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same

images, through the same scoring function. On the released crops this detector ranks third of seven: Sauvola local thresholding reaches 0.5145 and Sato ridge 0.4985 against our 0.4958, and the top three are separated by 0.0186 F1 — about the width of a gap this project has elsewhere called irresolvable at this sample size. The honest statement there is that this detector and two classical filters are indistinguishable.

On the whole fields the rest of this paper reports, it ranks fifth of seven, at 0.3444 against the best arm's 0.4697, and fifth by a width-blind skeleton F1. That is a worse standing than the crops show, and it is the standing on the set the rest of the paper argues from. The crops flatter every arm equally, but they do not flatter them by equal amounts: the classical ridge filters lose little between the two framings and this detector loses 0.1514 F1.

We report the comparison because a reader can reproduce it in an afternoon, and because the premise is worth more than the ranking: none of these seven arms is trained, all are tuned identically, and the figure shows what the whole classical family reaches on a benchmark where the published learned state of the art is far higher.

3.5 The ranking is buyable, and we checked what it costs

We tried to raise this detector's rank. The attempt succeeded, and that is the finding.

Take the detector's own output and dilate it. Give the dilation one tuned parameter, chosen leave-one-image-out exactly as every other arm chooses its own. Nothing is added: a morphological dilation cannot find a crack the detector did not already mark, and it reads no new evidence from the image. On the released crops that mask scores 0.6307 F1 against the shipped 0.4958 — a gain of 0.1349, and first place, clear of every classical filter.

It is not an artefact of pixel F1 either. Reducing both masks to centrelines and scoring agreement within 5 px — a measure that cannot see how wide a mask is drawn — the dilated mask still ranks first, at 0.4002 against the shipped 0.3279.
Dilation closes gaps in a fragmented mask, and gap-closing moves a centreline score.

Two further arms were run before this one and did worse. Growing the proposal through a Sato ridge response, which at least reads the image, was beaten by the plain dilation at matched parameter count; and on whole fields, growing it through a different field's ridge response scored 0.4043 against the real field's 0.4065. A step whose output barely changes when you feed it the wrong image is not reading the image.

So the rank can be fixed. What it costs is the two numbers this tool exists to report:

	area fraction, reported / true	mean width, reported / true
shipped detector	1.30x	0.75x
dilated, ranks first	2.54x	2.18x

On whole fields, buying 0.1349 F1 takes the area fraction from 1.30 times the labelled value to 2.54 times it, and the mean width from 0.75x to 2.18x. A reader of this tool would be told that 2.54 times as much of the field had cracked as the annotator marked. The benchmark and the measurement move in opposite directions, so we did not ship it.

We think this is the most useful thing in this section. The ranking on this dataset is substantially a ranking of how wide each method draws, which is the same conclusion Section 4 reaches from the other direction, and it means a table of F1 scores on broad-brush labels should not be read as a table of detection ability.

3.6 Where the missing recall actually is

The shipped recall on whole fields is 0.4347 pooled over pixels — this one subsection pools, because the question is which stage loses which pixels, and a per-image

mean cannot be decomposed that way; the per-image figure quoted elsewhere in this paper is 0.4123, and the two must not be mixed. The missing recall is not behind a threshold. Open both gates of the proposal stage to their limits — the MAD cut down to 0.5 and the absolute darkness floor to zero, which between them mark 53.4% of the image at precision 0.0196 — and recall still reaches only 0.579. 42.1% of the labelled crack is not dark at any scale, so no setting of either gate can propose it. That is a ceiling on this proposal mechanism for this material, not an operating point.

It also refutes a fix carried over from our own corpus, where the absolute darkness floor accounted for nearly all of the proposal-stage recall loss. On this alloy that gate costs 0.0006 recall. The gate that matters here is the MAD cut, and opening it is a precision trade rather than free recall: precision falls from 0.3214 to 0.0196.

4 What the scores measure

This is the part we would most want another group to check.

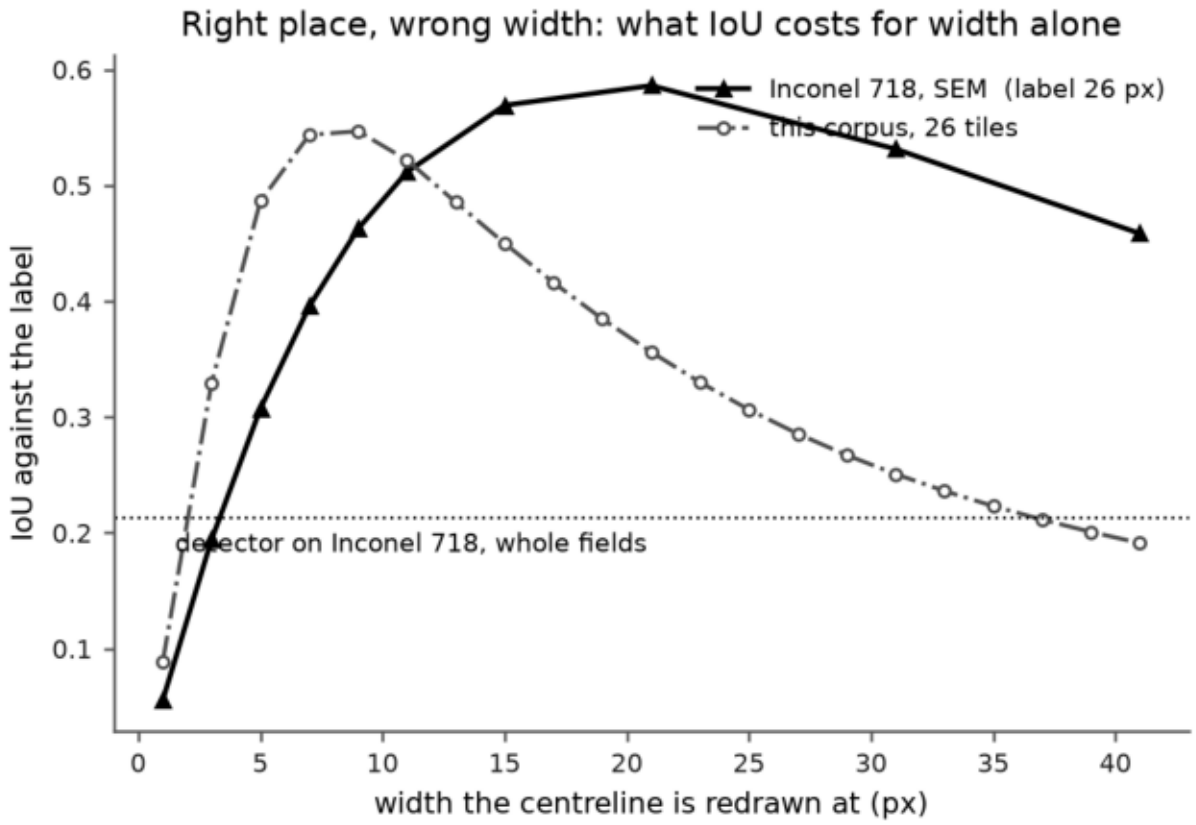


Figure 6. P3 - width sensitivity

Take the ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. Scored against the label it should be nearly perfect, and it is not: the best such reconstruction reaches IoU 0.5861. Most of the available IoU is being spent on agreeing about width, and the two curves in Figure 3 peak at different widths — 21 px for this dataset's labels against 9 px for our own corpus tiles — so the same detector's IoU can differ between corpora without its localisation differing at all.

This is not an upper bound, and we presented it as one. A real label varies in width along its length, so a detector that tracks that variation can score above a constant-width redraw of the same centreline. We had to be told that by the measurement.

And width does not explain the detector's difficulty, which we initially claimed it did. This dataset's labels are drawn at a median 25.8 px and its best constant-width reconstruction reaches 0.5861; our own corpus tiles, scored by the same construction, peak at 0.5467. Two corpora whose brush widths differ substantially reach comparable ceilings, so whatever makes these images hard for this detector, it is not how wide the labels are drawn.

That argument was stronger one draft ago. It rested on two SEM sets annotated four times apart in stroke width — 25.8 px and 6.4 px — reaching the same ceiling to within 0.007. The second of those was the EBC coating, withdrawn from this paper as a ceramic rather than a metal, and the argument is weaker without it. The measurement remains in docs/figures/sem_width_ceiling.json.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants singly and jointly, plus every morphology and size floor, under leave-one-specimen-out selection. No arm beat doing nothing on held-out data; the joint optimum lost 0.021 F1 when its selection was held out, and resolution-matching the scale ladder to the measured width ratio moved F1 by 0.0008.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of p = 0.125 — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

F1 is width-gameable where the labels are broad. On the same corpus a by-construction-correct 3 px crack scores F1 0.1260, below marking every pixel (0.1499), while F1 climbs to 0.64 on marked width alone. "Beats the no-information floor on F1" is necessary and not sufficient; precision lift separates where F1 does not.

Six precision rules validated on crops were deleters on frames. See §2.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the score in the units of the answer. F1 and IoU discard every pixel correctly left alone — 98% of the evidence on a 1.80%-crack image — and neither can state the error a user suffers, which is an area fraction off by some factor. Report labelled %, marked %, and their ratio; keep F1 and IoU beside them for comparability with the literature. The ratio is 1.30x here and 76.65x on a set this detector fails, and no F1 in this paper distinguishes those two situations as plainly.
- Never quote plain accuracy at low prevalence. It is 97.42% here, and marking nothing at all scores 98.20%. Balanced accuracy and MCC use the true negatives without being swamped by them.
- Report the no-information floor beside every score. Marking every pixel scores F1 0.0351 on this dataset. A bare score cannot be read without it, and across the sets measured for this project the floor moves by more than a factor of seven.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Our laboratory corpus detects about +0.14 F1 for a new method; we reported smaller differences for weeks before measuring this.
- Controls must be able to fail. Every control we built was broken on purpose to check

it noticed; two did not and were rewritten.

- Check the mask encoding and the bit depth before scoring anything. This set's masks are anti-aliased rather than two-valued and its images are 16-bit; an ordinary 8-bit conversion drove recall to 0.0012 and looked like a detector failure rather than a loading bug.
- Score whole fields, not crops of them. Cropping these images around their own cracks raises F1 by 0.1514 with no change to the method.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- The evidence is 31 images of one alloy, from a single released collection whose record is one sentence long and names no pixel size, no microscope and no annotation protocol. One crack class out of eighteen targets. That is enough to place the detector and not enough to characterise it, and it is the single largest weakness of this draft.
- A second class from the same collection was measured and removed. The EBC environmental barrier coating (24 fields, 0.26% prevalence) is a ceramic coating rather than a metal and is outside the application. The detector failed on it — 0.0102 precision at 0.6918 recall — and that failure is not reported above. Removing it also cost Section 4 its strongest form, which compared two sets annotated four times apart in brush width. Both measurements remain in docs/figures/.
- What was searched for and not found. Public crack segmentation data in this modality is close to non-existent. Ruled out, each for a stated reason: road and pavement sets (CrackForest, DeepCrack, Crack500, CrackSeg9k and the aggregated collections) are the wrong modality; MetalDAM is SEM steel with pixel masks but its five classes are phases and a generic "defect", not cracks; NASA's MicroNet superalloy tasks label precipitates, not cracks; a released Ti-6Al-4V fractography set (SE and BSE, 16 GB) ships images with no masks at all; and the published nickel-superalloy fatigue-crack segmentation work we found is tomography with no data release. A second independent SEM crack set would change this paper more than any further work on the method.
- Pavement results were measured and removed. The detector reaches F1 0.7932 and IoU 0.6809 on DeepCrack, ranking first of seven there. It is not reported above because a road-surface benchmark says nothing about micrographs, and the contrast between that ranking and the third-of-seven on SEM is better read as a warning about cross-modality benchmarking than as a result. The numbers remain in docs/figures/public_metrics.json.
- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.
- The corpus series in Figure 3 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.
- The detector's best score in this project is not on cracks. On MetalDAM's voids — compact, dark, high-contrast objects in SEM steel — it reaches MCC 0.5868 against 0.3402 on cracks in the superalloy, and reports an area fraction 0.74x the truth against 1.30x. That is what a multi-scale darkness threshold should do, and it bounds how much of this paper is about cracks specifically rather than about dark thin objects. Appendix A gives the full span.
- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Appendix A Every set measured, as a pixel budget

All five SEM sets this project has scored, in the units the tool reports. Two of them are not crack benchmarks and say so. Percentages are of all image pixels, averaged per image, mad_k chosen leave-one-image-out.

set	n	status	lab%	mark%	ok%	bad%	rep/true	bal.acc	MCC
Inconel 718, fields	31	the paper's benchmark	1.80%	2.34%	0.78%	1.56%	1.30x	69.82%	0.3402
Inconel 718, crops	26	same images, cut on cracks	4.53%	3.53%	2.11%	1.41%	0.78x	72.09%	0.4840
EBC coating, fields	24	withdrawn: ceramic	0.26%	20.29%	0.19%	20.10%	76.65x	74.52%	0.0662
EBC coating, crops	8	withdrawn	0.61%	19.70%	0.49%	19.21%	32.54x	76.72%	0.1108
MetalDAM voids	16	NOT cracks: voids	1.95%	1.44%	1.35%	0.09%	0.74x	78.03%	0.5868

The span is the point. The same detector, unchanged, reports an area fraction that is 0.74 times the truth on compact voids in steel and 76.65 times the truth on a coating at 0.26% prevalence. On the alloy it is 1.30x. Anyone quoting a single accuracy figure for this detector is quoting one point on that range.

MetalDAM is the most revealing row and it is not a crack result. Its targets are voids: compact, dark, high-contrast. The detector reaches MCC 0.5868 there against 0.3402 on cracks in the superalloy — its best score in this project is on the objects that are easiest to be dark. That is consistent with what it is: a multi-scale darkness threshold. Cracks are hard for it because they are thin and low in contrast, not because they are cracks.

The two pavement sets are absent. They were measured on P/R/F1/IoU before being removed from this paper as the wrong modality, and their downloads were not retained, so no budget was computed for them.

Appendix B. The detector, end to end



Figure 7. Every constant read from analysis/detector_overrides.py when the plate was drawn.